INCH-POUND

MIL-C-49137/7A(ER)
23 April 1992
SUPERSEDING
MIL-C-49137/7(ER)
25 November 1981

MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, ELECTROLYTIC (SOLID ELECTROLYTE), TANTALUM,
NON-POLAR, NONHERMETICALLY SEALED,
STYLES CX07 AND CX17

This specification is approved for use by the Electronics Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the capacitors described herein shall consist of this specification and the latest issue of MIL-C-49137.

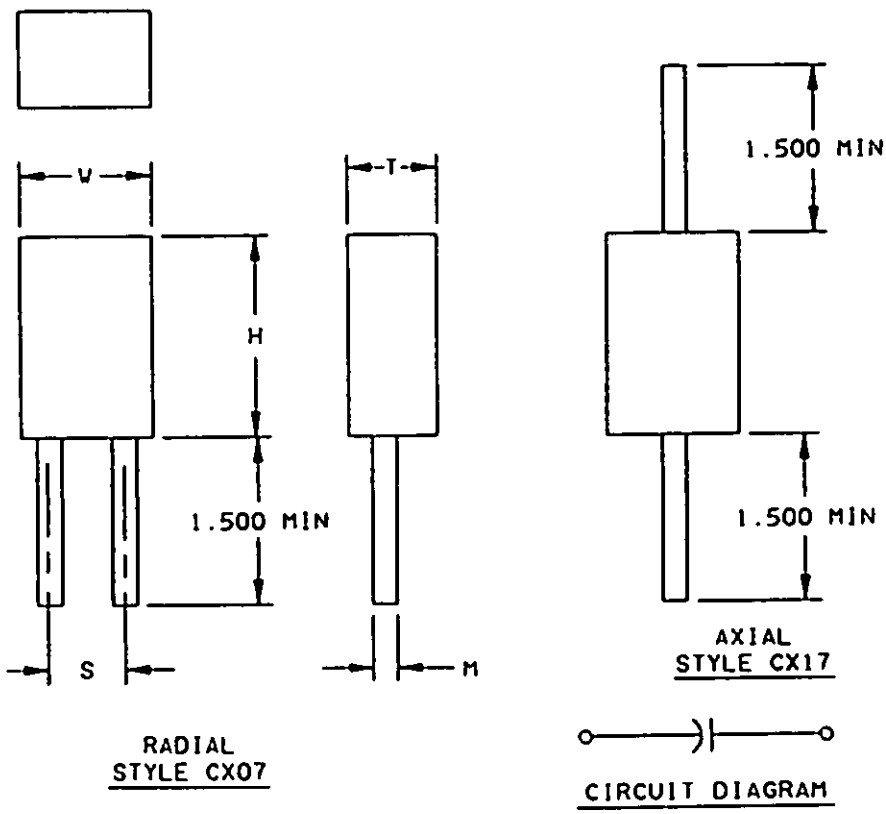


FIGURE 1. Styles CX07 and CX17 capacitors.

(A) Denotes changes

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:50Z
Check the source to verify that this is the current version before use.

Case size*	T		W		H		M	S
	Min	Max	Min	Max	Min	Max	±.002	
				CX07				
A	.090	.125	.090	.125	.145	.180	.010	.100 ±.020
B	.145	.180	.105	.140	.205	.240	.010	.100 ±.020
C	.190	.220	.150	.180	.285	.315	.016	.150 ±.025
D	.200	.230	.200	.230	.305	.335	.016	.200 ±.025
				CX17				
E	.090	.125	.090	.125	.185	.220	.010	.100 ±.020
F	.145	.180	.105	.140	.245	.280	.010	.100 ±.020
G	.190	.220	.150	.180	.340	.370	.016	.150 ±.025
H	.200	.230	.200	.130	.250	.390	.016	.200 ±.025

Inches	mm	Inches	mm	Inches	mm	Inches	mm
.002	.05	.125	3.18	.220	5.59	.335	8.47
.010	.25	.140	3.56	.230	5.84	.340	8.64
.016	.38	.145	3.68	.240	6.10	.350	8.89
.020	.41	.150	3.81	.245	6.22	.370	9.40
.025	.51	.180	4.57	.280	7.11	.390	9.91
.090	2.29	.190	4.83	.285	7.24	1.500	38.10
.100	2.54	.200	5.08	.305	7.75		
.105	2.67	.205	5.21	3.15	7.91		

NOTES:
1. Dimensions are in Inches.
2. Metric equivalents are given for general information only.

FIGURE 1. Styles CX07 and CX17 capacitors - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:50Z
Check the source to verify that this is the current version before use.

TABLE I. Styles CX07 and CX17 characteristics.

Type designation 1/	DC rated voltage (85°C)	Capacitance (nom)	Dissipation factor (max)	DC leakage (25°C) (max)	Case size	
					CX07	CX17
	Volts	µF	Percent	µA		
CX-7A475	2	4.7	10	0.5	A	E
CX-7B335	3	3.3	10	0.5	A	E
CX-7B106	3	10.0	10	0.5	B	F
CX-7C225	4	2.2	8	0.5	A	E
CX-7C685	"	6.8	"	1.0	B	F
CX-7C226	"	22.0	"	2.0	C	G
CX-7C336	"	33.0	"	3.0	D	H
CX-7D155	6	1.5	6	0.5	A	E
CX-7D475	"	4.7	"	1.0	B	F
CX-7D156	"	15.0	"	2.0	C	G
CX-7D226	"	22.0	"	3.0	D	H
CX-7F105	10	1.0	6	0.5	A	E
CX-7F335	"	3.3	"	1.0	B	F
CX-7F106	"	10.0	"	2.0	C	G
CX-7F156	"	15.0	"	3.0	D	H
CX-7H684	15	0.68	6	0.5	A	E
CX-7H685	15	6.8	6	2.0	C	G
CX-7H106	15	10.0	6	3.0	D	H
CX-7J474	20	0.47	6	0.5	A	E
CX-7J155	"	1.5	"	1.0	B	F
CX-7J225	"	2.2	"	1.0	B	F
CX-7J475	"	4.7	"	2.0	C	G
CX-7J685	"	6.8	"	3.0	D	H
CX-7K334	25	0.33	6	0.5	A	E
CX-7K105	"	1.0	"	1.0	B	F
CX-7K335	"	3.3	"	2.0	C	G
CX-7K475	"	4.7	"	3.0	D	H
CX-7M104	35	0.10	6	0.5	A	E
CX-7M154	"	0.15	"	"	"	"
CX-7M224	"	0.22	"	"	"	"
CX-7M334	"	0.33	"	"	"	"
CX-7M474	"	0.47	"	1.0	B	F
CX-7M684	"	0.68	"	1.0	B	F
CX-7M105	"	1.0	"	2.0	C	G
CX-7M155	"	1.5	"	2.0	C	G
CX-7M225	"	2.2	"	2.0	C	G
CX-7M335	"	3.3	"	3.0	D	H

1/ Complete type designation will include additional symbols to indicate style and capacitance tolerance.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:50Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

(A) Qualification not required.

Design and construction:

Dimensions and configuration: See figure I.

(A) Case material: Polyester sleeve and epoxy end filled or plastic encapsulated.

Terminals: Solder-coated metal.

DC rated voltage: See table I.

Operating temperature: -55°C to +85°C.

Construction analysis: Not applicable.

DC leakage (DCL): See table I.

Capacitance (cap.): See table I.

Cap. tolerance: ±10% (K) or ±20% (M).

Dissipation factor (DF): See table I.

Shock (specified pulse): Not applicable.

Terminal strength: Method 211 of MIL-STD-202.

Test condition A: 1 pound for case sizes A, B, E, AND F; 2 pounds for case sizes C, D, G, and H.

Test condition C: 1/2 pound applied force - 3 bends.

Stabililty at low and high temperatures: In accordance with MIL-C-49137.

Step 1 (+25°C):

DCL: See table I.

Cap: Within tolerance of table I value.

DF: See table I.

Step 2 (-55°C):

ΔCap.: With -15 percent of step 1 measured value.

DF: 200 percent of table I value.

Step 3 (+25°C):

DCL: see table I.

ΔCap.: Within 10 percent of step 1 measured value.

DF: See table I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:50Z
Check the source to verify that this is the current version before use.

Step 4 (+85°C):

DCL: 10 times 25°C limit (see table I).

ΔCap.: Within +15 percent of step 1 measured value.

DF: 150 percent of table I value.

Step 5 (+25°C):

DCL: See table I.

ΔCap.: Within 10 percent of step 1 measured value.

DF: See table I.

Surge voltage: In accordance with MIL-C-49137.

DCL: See table I.

ΔCap.: Within 10 percent of initial measured value.

DF: See table I.

Life: Method 108 of MIL-STD-202, test condition D (1,000 hours):

At 25°C:

DCL: 200 percent of table I value.

ΔCap.: Within 10 percent of initial measured value.

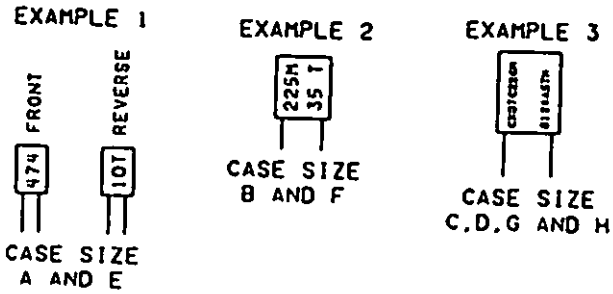
DF: See table I.

(A) Resistance to solvents: Not applicable.

Marking: In accordance with MIL-C-49137 and the following:
Case sizes A and E shall be marked as shown in example 1. The front shall be marked with the capacitance value coded in picofarads and the reverse shall be marked with the dc rated voltage and the manufacturer's trademark.

Case sizes B and F shall be marked as shown in example 2 (see MIL-C-49137).

Case sizes C, D, G, and H shall be marked as shown in example 3. The first line shall contain the type designation and the second line shall contain the date code (year and week), lot code, and manufacturer's trademark.



Voltage groups (85°C rated voltage):

I - 2 to 10 V inclusive
II - 15 to 35 V inclusive

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:50Z
Check the source to verify that this is the current version before use.

CONCLUDING MATERIAL

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5910-A461)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:50Z
Check the source to verify that this is the current version before use.